
Education

Cornell University, Ithaca, NY

Bowers College of Computing and Information Science

Doctor of Philosophy in Statistics and Data Science

Advisor: David S. Matteson

Cumulative GPA: 4.0

Aug. 2021 – Present

Cornell University, Ithaca, NY

College of Agriculture and Life Sciences

Masters of Science in Animal Science

Special Committee: Quirine M. Ketterings, Joe Guinness, Jan van Aardt

Cumulative GPA: 4.0

May 2021

Cornell University, Ithaca, NY

College of Arts and Sciences

Bachelor of Arts in Statistical Science and Minor in Computer Science

Dean's list (Fall 2016, Spring 2018)

May 2018

Research Interest

- Bayesian Time-Series Modeling (state-space models, structural time series)
- Volatility Modeling/ Uncertainty Quantification under Heteroskedasticity
- Spatial Statistics

Academic Fellowship / Awards

- 2023 PiTech Siegel PhD Impact Fellowship
- 2024 CALS Outstanding Teaching Assistant Awards in 2024

Publications

1. Cho J.B., Matteson D.S. Improving Conditional Coverage in Time-Series Foundation Models via Direct Volatility Modeling. ICLR 2026 TSALM Workshop Poster. OpenReview. 2026. Available from: <https://openreview.net/forum?id=dZx0R4Cm8d>
2. Cho J.B., Matteson D.S. BASTION: A Bayesian Framework for Trend and Seasonality Decomposition. AISTATS 2026 Poster. 2026. Available from: <https://openreview.net/forum?id=F0i4ncPYeI>
3. Cho, J.B. and Matteson, D. S. Smoothing Variances Across Time: Adaptive Stochastic Volatility. Journal of Business & Economic Statistics 2025; DOI: <https://doi.org/10.1080/07350015.2025.2602846>;
4. Lee, H., Cho J.B., Matteson, D. S., Domingue B. W. Dynamic Bayesian Item Response Model with Decomposition (D-BIRD): Modeling Cohort and Individual Learning Over Time URL <https://arxiv.org/abs/2506.21723>. - **Accepted for NCME AIME-CON**
5. Cho, J.B., Serneels, S., & Matteson, D.S. Non-Fungible Token Transactions: Data and Challenges. Data Science in Science 2022; DOI: <https://doi.org/10.1080/26941899.2022.2151950>
6. Cho J.B., Guinness J., Kharel T, Maresma Á., Czymmek K. J., van Aardt J., Ketterings Q. M. Proposed Method for Statistical Analysis of On-Farm Single Strip Treatment Trials. Agronomy. 2021; 11(10):2042. DOI: <https://doi.org/10.3390/agronomy11102042>

7. Cho, J.B., Guinness, J., Kharel, T.P. et al. Spatial estimation methods for mapping corn silage and grain yield monitor data. *Precision Agric* 22, 1501–1520 (2021). DOI: <https://doi.org/10.1007/s11119-021-09793-z>
8. Sunoj S, Cho J, Guinness J, van Aardt J, Czymmek KJ, Ketterings QM. Corn Grain Yield Prediction and Mapping from Unmanned Aerial System (UAS) Multispectral Imagery. *Remote Sensing*. 2021; 13(19):3948. DOI: <https://doi.org/10.3390/rs13193948>
9. Sunoj, S, Kharel, D, Kharel, T, Cho, J, Czymmek, KJ, Ketterings, QM. Impact of headland area on whole field and farm corn silage and grain yield. *Agronomy Journal*. 2021; 113: 147– 158. DOI: <https://doi.org/10.1002/agj2.20489>

Conference Presentations

1. Cho, J., Matteson, D.S., (2025) Smoothing Variances Across Time: Adaptive Stochastic Volatility. 2025 NBER-NSF Time-Series Conference <https://drive.google.com/file/d/1mX9NbTgh0dBkZH2iF6tE7FNBKpc324D-/view>
2. Cho, J.,Tupper, L.L., Curtis. T.M., Detweiler, M.A., Keese, C., Matteson, D.S., (2025) Robust Feature Construction for ECIS-Based Cell Type Classification Across Experiments 2025 ISU-NISS Conference on Artificial Intelligence and Statistics
3. Cho, J., Matteson. D.S., (2024) Locally Adaptive Random Walk Stochastic Volatility. NISS Graduate Student Network Research Conference 2024. <https://www.niss.org/events/niss-graduate-student-network-research-conference-2024>
4. Cho, J., Matteson. D.S., (2024) Locally Adaptive Random Walk Stochastic Volatility. The UPSTAT 2024 Conference. <https://community.amstat.org/rochester/events/upstats-2024/awards-and-prizes> - **Best poster presentation**
5. Cho. J., Serneels. S., & Matteson. D.S., (2023) Quantifying the Risk and Return of Non-Fungible Tokens. Conference on Advances in Time Series Analysis, Chicago, IL. <https://www.chicagobooth.edu/events/conference-on-advances-in-time-series-analysis>

Research Assistantship

Cornell Nutrient Management Spear Program, Ithaca, NY

Research Assistant – under Professor Quirine Ketterings

- Developed software for interpolating spatially correlated data (yield monitor data) using R
- Performed statistical analysis to understand the effect of nitrogen on corn yield

Dec. 2018 – May. 2021

Cornell Tech Future Automation Research Laboratory, New York, NY

Part-time Research Assistant – under Professor Wendy Ju

- Assisted data collection and organization, literature review and overall project management

Jul. 2018 – May. 2019

Cornell Statistical Consulting Unit, Ithaca, NY

Data Analyst

- Performed data analysis for researchers around Cornell. Tasks include dimensionality reduction, imputations, outlier detection, causal inference and clustering

Jul. 2018 – Dec. 2018

Teaching Assistantship

Graduate Teaching Assistant

STSCI 2150: Introductory Statistics for Biology

STSCI 4100: Multivariate Analysis

STSCI 4090: Linear Models with Matrices

ORIE 3500: Engineering Probability and Statistics II

Aug. 2019 – Present